MILITARY SPECIFICATION

FUZE, M509A1, POWER SUPPLY FOR AND GROUNDING SLEEVE ASSEMBLY FOR, PARTS FOR, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This Military Specification covers parts for assembling and packing of Fuze, M509A1, Power Supply and Grounding Sleeve Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation - Packaging, Methods Of
MIL-B-117	- Bag, Sleeve and Tubing - Interior Packaging
MIL-B-3106	- Board, Composition, Water-resistant, Solid (For Filler or Cushioning Pads)
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-I-45607	- Inspection Equipment, Supply and Maintenance of
MIL-F-50449	- Filler, Sheet Form (For Use In Ammunition Containers)
MIL-P-60312	- Parts, Molded, Plastic Foam, Polystyrene (For Use With Ammunition)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

FEDERAL

PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock, (Container Grade), and Cut Shapes
PPP-B-601	- Box, Wood, Cleated Plywood
PPP-B-636	- Box, Shipping, Fiberboard
PPP-T-76	- Tape, Pressure - sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)

STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing And Marking for Shipment of Inert Ammunition Components

DRAWINGS

PICATINNY ARSENAL

PRODUCT AND PACKAGE DRAWINGS

9318154	- Grounding Sleeve Assembly
9321944	- Power Supply, Fuze, M509A1
9283141	- Projectile, 105MM, HEAT-T, M456A1E2, Loading Assembly
9276311	- Cartridge, 105MM, HEAT-T, M456A1E2, Retrofit Assembly

3. REQUIREMENTS

3.1 General. -Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Power Supply.-The power supply shall comply with all requirements specified on Drawing (dwg.) 9321944 and with all requirements specified in applicable specifications.

3.3 Grounding Sleeve Assembly.-The grounding sleeve assembly shall comply with all requirements specified on dwg. 9318154 and with all requirements specified in applicable specifications.

3.4 Workmanship.-All parts and assemblies shall be free of chips, dirt, grease, rust, burrs, cracks and other foreign material. The cleaning methods used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

3.5 Piezoid.

3.5.1 The piezoid shall be capable of passing all required electrical tests after temperature cycling, as well as at ambient temperatures.

3.5.2 The piezoid shall not exhibit any cracking when subjected to a minimum (min.) compressive load of 50,000 psi.

3.6 Proving Ground.-The applicable loaded projectile shall comply with the following requirements when assembled with a power supply and tested as specified in 4.5.5.

3.6.1 Graze Functioning.-The projectile shall detonate "high-order" on initial impact with a gravel pad located 1000 ± 200 meters from the gun muzzle.

3.6.2 Safety.-There shall be no premature bursts, in flight, prior to impact.

3.7 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Resistor.-The resistor shall not exhibit any cracking when subjected to a min. compressive load of 10,000 psi.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Power Supply, Fuze, M509A1	9321944	30
Piezoid	9276161	25
Spacer Terminal Assembly	9276272	5
Anvil Terminal Assembly	9321827	5
Grounding Sleeve Assembly	9318154	30
Resistor	9321946	25

In addition, the submission shall include 10 each of every component and every subassembly not listed above. (Where a multiple cavity mold is used in the manufacturing process, a minimum of 5 samples shall be submitted from each mold cavity. Each part shall be identified with its mold cavity).

4.3.2 Inspections to be Performed.-The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to all examinations and tests requirements of the applicable drawings. After inspection, each power supply shall be cut open, and the piezoid and resistor shall be subjected to all applicable electrical tests specified in 4.4.2 of this specification.

4.3.3 Rejection.-See MIL-A-48078 (PA).

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of power supplies shall contain piezoid lots produced from no more than a single batch of composition mix (See 6.6).

4.4.2 Examination.-See MIL-A-48078

a. Sampling Plans.-Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48753(PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Holder, Switch Terminal	1 OF 1	9276157
			NEXT HIGHER ASSEMBLY 9276156

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131	Any feature or operation missing		0.40%	3.2	Visual
132	Thickness, Min		0.40%	3.2	Gage
133	Thickness of crimp lip		0.40%	3.2	Gage
134	Outside diameter, min.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

Source: https://assist.dla.mil Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48753(PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Pin, Switch Terminal		9276158
			NEXT HIGHER ASSEMBLY 9276156

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
101	Length of effective thread, min		0.40%	3.2	Gage
102	Overall length		0.40%	3.2	Gage
103	Outside diameter, min.		0.40%	3.2	Gage
Minor					
201	Thread pitch diameter, min.		0.65%	3.2	Gage
202	Thread major diameter, min.		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

Source: dlist.dla.mil Downloaded: 2026-10-22T19:38Z Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48753(PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Switch Terminal Assembly	1 OF 1	9276156
			NEXT HIGHER ASSEMBLY 9276272

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Leak test of glass seal		4.4.3.1	3.2	Test
102	Pull test of pin		4.4.3.1	3.2	Test
103	Perpendicularity		0.40%	3.2	Gage
104	True position		0.40%	3.2	Gage
Major B					
131	Protrusion length of pin		0.40%	3.2	Gage
Minor					
201	Evidence of glass on threads		0.65%	3.2	Visual
202	Protective finish missing or incomplete		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

Check the source to verify that this is the current version before use.
https://assist.dla.mil - Downloaded: 2026-10-22T19:38Z

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Insulator, Spacer	1 OF 1	9276275
			NEXT HIGHER ASSEMBLY 9276272

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131	Thickness, min.		0.40%	3.2	Gage
132	Parallelism		0.40%	3.2	Gage
Minor					
101	Outside diameter, min.		0.65%	3.2	Gage
102	Diameter of hole, max.		0.65%	3.2	Gage
103	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

Source: http://assist.daps.dla.mil - Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48753(PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Spacer		9276276
			NEXT HIGHER ASSEMBLY 9276272

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131	Outside diameter, min.		0.40%	3.2	Gage
132	Thickness, min		0.40%	3.2	Gage
133	True position		0.40%	3.2	Gage
Minor					
201	Thread pitch diameter, max.		0.65%	3.2	Gage
202	Thread minor diameter, max.		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.4	Visual
204	Protective finish missing or incomplete		0.65%	3.2	Visual

NOTES:

Source: http://assist.daps.dla.mil - Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.6	Spacer Terminal Assembly	1	1	9276272
				NEXT HIGHER ASSEMBLY 9321944

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	True position of spacer and spacer insulator		0.40%	3.2	Gage
102	Height, max.		0.40%	3.2	Gage
Major B					
131	Any part missing or improperly assembled		100%	3.2	Visual
132	Visible gaps present between component parts		0.40%	3.2	Visual
133	Assembly loose		0.40%	3.2	Manual
Minor	None defined				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Holder, Spike Terminal		9276153
			NEXT HIGHER ASSEMBLY 9322040

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131	Any feature or operation missing		0.40%	3.2	Visual
132	Thickness, min		0.40%	3.2	Gage
133	Thickness of crimp lip		0.40%	3.2	Gage
134	Outside diameter, min.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48753(PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Pin, Spike Terminal	1 OF 1	9321959
			NEXT HIGHER ASSEMBLY 9322040

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131	Length of effective thread, min		0.40%	3.2	Gage
132	Overall length, min.		0.40%	3.2	Gage
133	Outside diameter		0.40%	3.2	Gage
Minor					
201	Thread pitch diameter, min.		0.65%	3.2	Gage
202	Thread major diameter, min.		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.4	Visual

Source: https://assist.dla.mil - Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48753 (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Spike Terminal Assembly	1 OF 1	9322040
			NEXT HIGHER ASSEMBLY 9321827

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Leak test of glass seal		4.4.3.1	3.2	Test
102	Pull test of pin		4.4.3.2	3.2	Test
103	Perpendicularity		0.40%	3.2	Gage
104	True position		0.40%	3.2	Gage
Major B					
131	Protrusion length of pin		0.40%	3.2	Gage
Minor					
201	Evidence of glass on threads		0.65%	3.2	Visual
202	Protective finish missing or incomplete		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48753 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Insulator, Envelope		9321826
			NEXT HIGHER ASSEMBLY
			9321827

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness, min (check through major and minor diameters and through shoulder)		0.40%	3.2	Gage
102	Major inside diameter, max		0.40%	3.2	Gage
103	Minor inside diameter, max		0.40%	3.2	Gage
104	Depth to inside shoulder		0.40%	3.2	Gage
105	Depth of minor diameter		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Resistor				9321946
					NEXT HIGHER ASSEMBLY
					9321827
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Physical and electrical properties at +125°F		4.4.3.6	3.2,3.8	4.5.6
102	Physical and electrical properties at -35°F		4.4.3.6	3.2,3.8	4.5.6
Minor					
201	Thickness, min		0.65%	3.2	Gage
202	Outside diameter		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

Source: https://assist.dla.mil Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9321952
4.4.2.12	Anvil				NEXT HIGHER ASSEMBLY 9321827
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	True position of threads		0.40%	3.2	Gage
102	True position of c'bore diameter		0.40%	3.2	Gage
103	True position of c'bore boss		0.40%	3.2	Gage
104	Depth at c'bore insulator groove		0.40%	3.2	Gage
105	Parallelism of rear face		0.40%	3.2	Gage
106	Flatness of c'bore boss face		0.40%	3.2	Gage
Major B					
131	Outside diameter, min.		0.40%	3.2	Gage
132	C'bore diameter, max.		0.40%	3.2	Gage
133	C'bore boss diameter, min.		0.40%	3.2	Gage
134	Thickness through c'bore boss		0.40%	3.2	Gage
135	Depth to c'bore boss		0.40%	3.2	Gage
136	Outside diameter of shoulder, min.		0.40%	3.2	Gage
137	Thickness of shoulder		0.40%	3.2	Gage
138	Depth of insulator groove in shoulder		0.40%	3.2	Gage
139	True position of insulator groove in shoulder		0.40%	3.2	Gage
Minor					
201	Thread pitch diameter, max.		0.65%	3.2	Gage
202	Thread minor diameter, max.		0.65%	3.2	Gage
203	Surface finishes incorrect		0.65%	3.2	Visual/Comparison Blocks
204	Protective finish missing or incomplete		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2025-10-22T19:39Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48753(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9321827
4.4.2.13	Anvil Terminal Assembly				NEXT HIGHER ASSEMBLY 9321944
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
101	Any part missing		100%	3.2	Visual
102	Visible gap present between resistor and spike terminal assembly		0.40%	3.2	Visual
103	Dimension from face of terminal to shoulder of anvil, max.		0.40%	3.2	Gage
Minor	None defined				

NOTES:

Source: https://quicksearch.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Piezoid	1 OF 1	9276161
			NEXT HIGHER ASSEMBLY: 9321944

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Parallelism of faces		0.40%	3.2	Gage
102	Electrical resistance of silvered surface (both sides)		0.40%	3.2	Test
103	Open circuit voltage after release of compressive load, min.		100%	3.2	Test
104	Capacitance, min.		100%	3.2	Test
105	Electrode to electrode electrical resistance, min.		100%	3.2	Test
106	Minimum electrical characteristics after temperature cycle		4.4.3.3	3.5.1	4.5.2
107	Minimum compressive strength		4.4.3.4	3.5.2	4.5.3
108	Silver coating extending over edge of piezoid		100%	3.2	Visual
Major B					
131	Outside diameter, min.		0.40%	3.2	Gage
132	Thickness		0.40%	3.2	Gage
133	Absence of void in paint marking positive face		0.40%	3.2	Gage
134	Surface finish incorrect		0.40%	3.2	Visual/Comparison Blocks
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2022-10-22T14:14:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9276164
4.4.2.15	Insulator, Piezoid				NEXT HIGHER ASSEMBLY 9321944
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	True position of inside diameter		0.40%	3.2	Gage
102	True position of outside lip diameter		0.40%	3.2	Gage
103	True position of inside lip diameter		0.40%	3.2	Gage
Major B					
131	Inside diameter, max.		0.40%	3.2	Gage
132	Thickness of internal lip, max.		0.40%	3.2	Gage
133	Dimension to external lip, min.		0.40%	3.2	Gage
134	Height		0.40%	3.2	Gage
135	Outside diameter, min.		0.40%	3.2	Gage
136	Inside lip diameter, min.		0.40%	3.2	Gage
137	Outside lip diameter, max.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

Check the source to verify that this is the current version before use.
Downloaded: 2026-10-22T19:38Z

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48753(PA)

PARAGRAPH	TITLE		SHEET 1 of 1		DRAWING NUMBER 9276165
4.4.2.16	Plate, Fulcrum				NEXT HIGHER ASSEMBLY 9321944
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	True position of minor outside diameter		0.40%	3.2	Gage
102	True position of tang ramp		0.40%	3.2	Gage
103	True position of shorting bar locating surface		0.40%	3.2	Gage
Major B					
131	Dimension between shorting bar locating surfaces, max.		0.40%	3.2	Gage
132	Width of tang ramp, max.		0.40%	3.2	Gage
133	Thickness		0.40%	3.2	Gage
134	Thickness through shorting bar seating surface		0.40%	3.2	Gage
135	Height of minor diameter boss		0.40%	3.2	Gage
136	Depth of tang ramp at basic dimension, max.		0.40%	3.2	Gage
137	Minor outside diameter, min.		0.40%	3.2	Gage
138	Major outside diameter, min.		0.40%	3.2	Gage
139	Depth of relief slot		0.40%	3.2	Gage
140	Width of relief slot		0.40%	3.2	Gage
141	Location of relief slot		0.40%	3.2	Gage
Minor					
201	Protective finish incomplete or missing		0.65%	3.2	Visual
202	Surface finishes incorrect		0.65%	3.2	Visual/Comparison Blocks
203	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2022-12-27T11:38:37Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.17	Bar, Shorting	1 OF 1	9276166
			NEXT HIGHER ASSEMBLY: 9321944

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	True position of end of tang		0.40%	3.2	Gage
102	True position of root of tang		0.40%	3.2	Gage
103	True position of tang locating flats		0.40%	3.2	Gage
104	Deflection of tang under load		0.40%	3.2	Test
105	Hardness		0.40%	3.2	Test
106	Flatness		0.40%	3.2	Gage
107	Outside diameter, Min.		0.40%	3.2	Gage (Note 1)
108	Thickness		0.40%	3.2	
Major B					
131	Dimension between tang locating flats, min.		0.40%	3.2	Gage
132	Width of tang end, min.		0.40%	3.2	Gage
133	Width of root of tang, min.		0.40%	3.2	Gage
134	Dimension between tang locating flat and tang root		0.40%	3.2	Gage
135	Width of tang slot		0.40%	3.2	Gage
Minor					
201	Protective finish missing or incomplete		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.4	Visual

Source: http://assist.daps.dla.mil/quicksearch/ 2026-10-22 09:08Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.18	Envelope		9276172
			NEXT HIGHER ASSEMBLY
			9321944

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	True position of minor internal diameter		0.40%	3.2	Gage
102	True position of inside diameter at aft end		0.40%	3.2	Gage
Major B					
131	Wall thickness, min.		0.40%	3.2	Gage
132	Major internal diameter, max.		0.40%	3.2	Gage
133	Minor internal diameter, max.		0.40%	3.2	Gage
134	Depth		0.40%	3.2	Gage
135	Depth of minor internal diameter		0.40%	3.2	Gage
Minor					
201	Protective finish missing or incomplete		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-02-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48753(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9276246
4.4.2.19	Insulator, Switch Terminal				NEXT HIGHER ASSEMBLY 9321944
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
101	Thickness, min.		0.40%	3.2	Gage
102	Diameter of hole, max.		0.40%	3.2	Gage
Minor					
201	Outside diameter		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.4	Visual
203	True position		0.65%	3.2	Gage

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.20	Washer, Felt		9276278
			NEXT HIGHER ASSEMBLY
			9321944

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Thickness		0.65%	3.2	Gage
202	Outside diameter, min		0.65%	3.2	Gage
203	Diameter of hole, max		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.4	Visual
205	True position		0.65%	3.2	Gage

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.21	Spring, Contact		9276281
			NEXT HIGHER ASSEMBLY 9321944

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131	Small coil inside diameter, max.		0.40%	3.2	Gage
132	Free length, min.		0.40%	3.2	Gage
133	Wire diameter		0.40%	3.2	Gage
134	Radius missing on aft end of wire		0.40%	3.2	Visual
135	Load at solid height		0.40%	3.2	Test
Minor					
201	Spring not left hand coiled		0.65%	3.2	Visual
202	Number of active coils		0.65%	3.2	Visual

NOTES:

Source: https://dasi.cd.mil Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.22	Disc, Support		9318138
			NEXT HIGHER ASSEMBLY
			9321944

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131	Thickness, min		0.40%	3.2	Gage
132	Outside diameter, Min.		0.40%	3.2	Gage
133	Diameter of hole, max.		0.40%	3.2	Gage
Minor					
201	Outside diameter, max.		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.4	Visual
203	True position		0.65%	3.2	Gage

NOTES:

Source: https://assist.dla.mil Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9321944
4.4.2.23	Power Supply, Fuze, M509				NEXT HIGHER ASSEMBLY 9276311
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Safety*		4.4.3.5.2	3.6.2	Visual/Photographic
Major A					
101	Air leakage exceeds max allowable rate		100%	3.2	Test (Note 1)
102	Electrical resistance, terminal-to-terminal, under min. allowable		100%	3.2	Test
103	Electrical resistance, switch terminal to envelope, under min. allowable		100%	3.2	Test
104	Electrical resistance, spike terminal-to-switch terminal, under min. allowable		100%	3.2	Test
105	Capacitance, terminal-to-terminal, under min. allowable		100%	3.2	Test
106	Shorting bar missing or improperly assembled		100%	3.2	Visual
107	Graze functioning*		4.4.3.5.1	3.6.1	4.5.5.1
108	Electrical resitance, spike terminal-to envelope, under min. allowable		100%	3.2	Test
Major B					
131	Piezoid insulator missing		100%	3.2	Visual
132	Positive face of piezoid not facing spike terminal		100%	3.2	Visual
133	Solder flux on either terminal assembly, discoloration of solder or of plating in the immediate area of the solder joints		0.40%	3.2	Visual

NOTES: Note 1 - This test shall be performed prior to all electrical tests.
*Proving ground

Source: https://assist.dla.mil -- Downloaded: 2026-10-22 10:18Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.23	Power Supply, Fuze, M509				9321944
					NEXT HIGHER ASSEMBLY
					9276311
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major B					
135	Piezoid insulator above flush with Anvil		0.40%	3.2	Gage
136	Presence of foreign material on insulators prior to or after assembly		100%	3.2	Visual
Minor					
201	Switch terminal insulator or contact spring missing or insecure		0.65%	3.2	Manual/Visual
202	Evidence of poor workmanship		0.65%	3.4	Visual
203	Marking missing, incomplete, misleading or unidentifiable		0.65%	3.2	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Insulator, Grounding Sleeve	1 OF 1	9318152
			NEXT HIGHER ASSEMBLY 9318154

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Outside diameter		0.40%	3.3	Gage
102	Diameter of snap - lip, min.		0.40%	3.3	Gage
Major B					
131	Thickness of face		0.40%	3.3	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual
202	Inside diameter		0.65%	3.3	Gage

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9318153
4.4.2.25	Sleeve, Grounding				NEXT HIGHER ASSEMBLY 9318154
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131	Pitch diameter of external threads, min.		0.40%	3.3	Gage
132	Major diameter of external threads, min.		0.40%	3.3	Gage
133	Pitch diameter of internal threads, max.		0.40%	3.3	Gage
134	Minor diameter of internal threads, max.		0.40%	3.3	Gage
135	Depth		0.40%	3.3	Gage
136	Depth to inside shoulder		0.40%	3.3	Gage
137	Major outside diameter		0.40%	3.3	Gage
138	Diameter of external undercut		0.40%	3.3	Gage
139	Diameter of internal undercut		0.40%	3.3	Gage
140	Inside diameter of power supply cavity		0.40%	3.3	Gage
141	Diameter of power supply terminal through hole, max.		0.40%	3.3	Gage
142	Height		0.40%	3.3	Gage
143	Depth to external shoulder		0.40%	3.3	Gage
144	Diameter of insulator snap - lip receiver, max.		0.40%	3.3	Gage

NOTES:

Source: https://assist.dla.mil - Downloaded: 2026-01-22T19:38Z
Check the source to verify that this is the current version before use.

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9318153
4.4.2.25	Sleeve, Grounding				NEXT HIGHER ASSEMBLY 9318154
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual
202	Pitch diameter of external threads, max.		0.65%	3.3	Gage
203	Major diameter of external threads, max.		0.65%	3.3	Gage
204	Pitch diameter of internal threads, min.		0.65%	3.3	Gage
205	Minor diameter of internal threads, min.		0.65%	3.3	Gage
206	True position of minor outside diameter, forward of threads		0.65%	3.3	Gage
207	True position of power supply terminal through hole		0.65%	3.3	Gage
208	True position of internal threads		0.65%	3.3	Gage

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9318154
4.4.2.26	Grounding Sleeve Assembly				NEXT HIGHER ASSEMBLY 9276311
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Height, max		0.40%	3.3	Gage
102	Presence of foreign material on insulator prior to assembly		0.40%	3.3	Visual
Minor					
201	Insulator insecure or missing		0.65%	3.3	Manual/Visual
202	Inside diameter through insulator, min		0.65%	3.3	Plug Gage (Note)

NOTES:
Plug gage shall be inserted through insulator from inside of assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS MIL-F-48753 (PA)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.27	Packing of Power Supply (See 5.1)		1 1		N/A
					NEXT HIGHER ASSEMBLY N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	One or more power supplies missing from any bottom support, prior to assembly of top support		0.65%	3.1	Visual
202	Packing loose, within fiber box		0.65%	3.1	Visual/Manual
203	Fiber box damaged, improperly sealed or marked		0.65%	3.1	Visual
204	Bag damaged, improperly sealed or marked or missing (Level B)		0.65%	3.1	Visual
205	Packing loose, within wood box (Level B)		0.65%	3.1	Visual/Manual
206	Wood box damaged, improperly sealed or marked (Level B)		0.65%	3.1	Visual
207	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.28	Packing of Grounding Sleeve (See 5.1)	1	1	N/A
				NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	One or more grounding sleeves missing from any layer		0.65%	3.1	Visual
202	Packing loose, within fiber box		0.65%	3.1	Visual/Manual
203	Fiber box damaged, improperly sealed or marked		0.65%	3.1	Visual
204	Bag damaged, improperly sealed or marked, or missing (Level B)		0.65%	3.1	Visual
205	Packing loose, within wood box (Level B)		0.65%	3.1	Visual/Manual
206	Wood box damaged, improperly sealed or marked (Level B)		0.65%	3.1	Visual
207	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

4.4.3 Testing

4.4.3.1 Leak Test of Glass Seal.-Thirty samples shall be selected from each lot for this test. If any sample fails to comply with the applicable requirement, the lot shall be rejected.

4.4.3.2 Pull Test of Pin.-Thirty samples shall be selected from each lot for this test. If any sample fails to comply with the applicable requirement, the lot shall be rejected.

4.4.3.3 Minimum Electrical Characteristics After Temperature Cycle.-Thirty piezoids shall be selected from each lot for this test. If any sample fails to comply with the applicable requirement, the lot shall be rejected.

4.4.3.4 Minimum Compressive Strength.-Ten piezoids shall be selected from each lot for this test. If any sample fails to comply with the applicable requirement, the lot shall be rejected. Samples subjected to min. compressive strength test shall not be returned to the lot.

4.4.3.5 Proving Ground

4.4.3.5.1 Graze Functioning.-A sample of twenty (20) power supplies shall be randomly selected from each lot for this test. Each sample power supply shall be assembled onto a cartridge comprised of components (projectile, fuze, shaped-charge liner, etc.) previously tested and accepted in accordance with applicable specifications. If six (6) or more cartridges fail to comply with the applicable requirement, the power supply lot shall be rejected. If one (1) or more projectiles fail to initially impact the gravel pad, the power supply lot shall be referred and results of such impacts shall be reported.

4.4.3.5.2 Safety.-All cartridges fired in proving ground testing shall be observed for premature bursts, in flight, prior to impact. If any power supply fails to comply with the applicable requirement, the lot shall be rejected.

4.4.3.6 Physical and Electrical Properties of Resistor at Temperature Extremes.-A sample of ten (10) resistors shall be randomly selected from each lot for each of these tests. If any resistor fails to comply with the applicable requirement, the lot shall be rejected. Sample resistors subjected to these tests shall not be returned to the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

4.5 Test Method and Procedures

4.5.1 Materials, Components and Processes.-Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and Mechanical Inspection.-The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein

4.5.2 Minimum Electrical Characteristics After Temperature Cycle.-Sample piezoids shall be temperature conditioned at +145°F ± 5°F (piezoid temperature) for one hour (min.) then conditioned at -65°F ± 5°F (piezoid temperature) for one hour (min.). The entire cycle shall be accomplished within a eight hour period. After temperature cycling, the piezoids will be conditioned at 70°F ± 5°F for 48 hours and subjected to each of the following tests:

a. Open circuit voltage after release of compressive load, min.

b. Capacitance, min.

c. Electrode to electrode electrical resistance, min.

Failure is defined as the inability of any sample piezoid to comply with the applicable requirement for any or all three tests.

4.5.3 Minimum Compressive Strength.-The compressive strength of the piezoid shall be tested with the sample piezoid between steel platens of Rockwell Hardness C54 minimum.

The following testing parameters are advisory:

a. Platen thickness one inch minimum

b. Length to diameter ratio 1:2

c. Platens to be flat to .0005 inches

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

d. Platens to have 8RMS finish

e. All testing surfaces to be parallel within .001 inches.

f. Platens and piezoid sample to be concentric to center of load and free of all foreign material

g. Constant loading rate of 10,000 pounds per minute

h. Compression loading of the piezoid to be continuous

4.5.4 Dimensional Control of Cast, Stamped, or Molded Parts.-In place of the normal sampling associated with the classification of Defects, and subject to approval of the Government procuring agency, the optional sampling plans in paragraphs 4.5.4.1, 4.5.4.2, and 4.5.4.3 may be used for dimensional control of cast, stamped, or molded parts. Use of these optional plans is limited to those dimensions and features controlled by the tool, die, or mold cavity. For all specified tests, visual examinations, and examinations of dimensions or features not developed exclusively by the tool, die, or mold cavity, these optional plans are not applicable. In the event of in-process reject(s) during optional sampling, corrective action applicable to parts produced prior to the rejected, and subsequent to the last acceptable sample, shall require approval of the Government procuring agency.

4.5.4.1 Cast Parts.-A sample of at least five (5) parts (as cast) from each cavity shall be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of the cavity during production. Inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of 5000 parts. Individual cavity identification must be provided. If any defective parts are found during qualification of the cavity, the cavity producing the defective part shall not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be remomed from production. Further, that portion of production since the last control check shall be returned for corrective action. All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test specified above.

4.5.4.2 Stamped Parts.-A sample of at least ten (10) parts shall be dimensionally inspected to qualify the tool

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production. If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production. If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned for corrective action. All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above.

4.5.4.3 Molded Parts.-A sample of five (5) parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be dimensionally inspected after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned for corrective action. All cavities removed from production because of some fault, may, after reworking, be returned to production, providing they pass the qualification test above.

4.5.5 Ballistic Test.-These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The sample cartridges shall be fired in a gun for which the cartridge is standard and observation shall be made for proper functioning and for premature bursts, in flight, prior to impact.

4.5.5.1 Graze Functioning.-The ammunition for this test shall be conditioned at a temperature of 70 Degrees Fahrenheit (°F) ±5 °F for 16 hours min. prior to firing. Test cartridges shall be fired for graze functioning at a gravel impact field, at a range of 1000 meters.

4.5.6 Physical and Electrical Properties of Resistor at Temperture Extremes.-The sample resistors shall be conditioned at the required temperature ± 5°F (10 at +125°F,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

10 at -35°F) for 8 continuous hours min. The resistors shall be removed from conditioning, and subjected to the following tests within 20 minutes max. after removal from conditioning:

a. Resistance
b. Power dissipation
c. Compressive loading.

5. Preparation for Delivery

5.1 Preservation and Packing (All dimensions are in inches)

5.1.1 Power Supply Level C.-Pack 36 power supplies, aft terminal end down, into the 36 cavities of the bottom support. Bottom support to be manufactured in accordance with Figure 1. Position top support (Ref Figure 2) over bottom support. Tape two halves together with 2 strips of 2 inch wide Spec PPP-T-76 tape. The two strips of tape to cross each other in center. Pack 5 layers (180 power supplies) in a Type CF, Class Domestic, Variety DW, grade 275, style RSC fiber box Spec PPP-B-636. The inside dimensions of fiber box to be 8 3/8 x 8 3/8 x 13 1/8 high. Add commercial chipboard fillers 7 3/4 x 7 3/4 x 1/16 thick or 7 3/4 x 13. x 1/16 thick as applicable on sides, ends and top as required for tight packing. Tape fiber box in accordance with Spec PPP-B-636 as applicable for fiber boxes overpacked or shipped as exterior containers.

5.1.2 Power Supply Level B.-In addition to fiber box specified in Level C, overpack fiber box in a Class E, Spec MIL-B-117 bag after blunting corners of box. Exhaust all excess air within bag and seal in accordance with MIL-P-116. Size of bag to be 18 1/2 min. open face x 26. min. deep. Pack 18 bagged fiber boxes (3 x 3 x 2 high) in an overseas type, grade B, cleated-plywood box, Spec PPP-B-601, any permissible style. Box to be provided with skids in accordance with Para. 3.3.3 of Spec. Requirements for box to be for 380 pounds weight of contents (Ref Table III). Inside dimensions of box to be 27 3/8 x 27 3/8 x 30. 1/4 high. Add filler material 26. square x 1/16 thick on sides, ends and top as required for tight packing. Filler material to comply with Spec MIL-B-3106 or MIL-F-50449.

5.1.3 Power Supply Level A.-Same as Level B.

5.1.4 Ground Sleeve Level C.-Pack 1620 grounding sleeve assemblies in 9 layers of 180 each in a partitioned Type CF, Class Domestic, Variety DW, grade 275, CSSC style fiber box,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

Spec PPP-B-636. The inside dimensions of fiber box to be 18 1/4 x 14 5/8 x 10 11/16 high. The partitions shall be of 12 x 15 pattern, with cell size 1 3/16 square x 1 1/16 high. Partitions shall slide fit into box. Material to be .032 thick, Type SF, Class Domestic, grade 125, Spec PPP-F-320. Place a partitionin box, pack 180 grounding sleeve assemblies, insulator end up. Repeat for 8 additional layers placing a separator beneath each layer. Separator to be Type CF, Class Domestic, Variety SW, grade 175, "B" Flute, size 18. x 14 3/8. Place additional separators atop contents as required for tight packing. Tape fiber box in accordance with Spec PPP-B-636 as applicable for fiber boxes overpacked or shipped as exterior containers.

5.1.5 Grounding Sleeve Level B.-In addition to fiber box specified in Level C, overpack fiber box in a Class E, Spec MIL-B-117 bag after blunting corners of box. Exhaust all excess air within bag and seal bag in accordance with Spec MIL-P-116. Size of bag to be 28. min open face x 34. min. deep. Pack 8 bagged fiber boxes (2 x 2 x 2 high) in an overseas type, grade B, cleated-plywood box, Spec PPP-B-601, any permissible style. Box to be provided with skids in accordance with Para. 3.3.3 of Spec. Requirements for box to be for 560 pound contents (Ref Table III). Inside dimensions of box to be 39. x 31. x 24 1/4 high. Add filler material 23 3/4 x 30 1/2 x 1/16 thick or 30 1/2 x 38 1/2 x 1/6 thick as applicable on ends, sides and top as required for tight packing. Filler material to comply with Spec MIL-B-3106 or MIL-F-50449.

5.1.6 Grounding Sleeve Level A.-Same as Level B.

5.2 Marking

5.2.1 Level C.-Marking of fiber box to be in accordance with MIL-STD-1169.

5.2.2 Level B.-Marking of fiber box, bag and box to be in accordance with Level A, MIL-STD-1169.

5.2.3 Level A.-Same as Level B.

6. NOTES

6.1 Intended Use.-The part covered by this specification is intended for use as part of the M509A1, PIBD, Fuzing System.

6.2 Ordering Data.-See MIL-A-48078 (PA).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

6.3 Submission of Designs for Approval.-See 6.2.3 of MIL-A-48078 (PA). Submit equipment designs, as required, to:

Commander
US Army Armament R&D Command
ATTN: DRDAR-QAT-I
Dover, New Jersey 07801

6.4 Inspection Lot Size.-It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of test or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5 Visual Examination.-When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.6 Batch.-A batch is defined as a quantity of piezoid ingredients blended in a single container at one time.

6.7 Proving Ground Test Summary.-

Test	Sample Size	Requirement	Test Procedures Paragraph
Graze Functioning	20 rds	3.6.1	4.5.5.1
Safety		3.6.2	

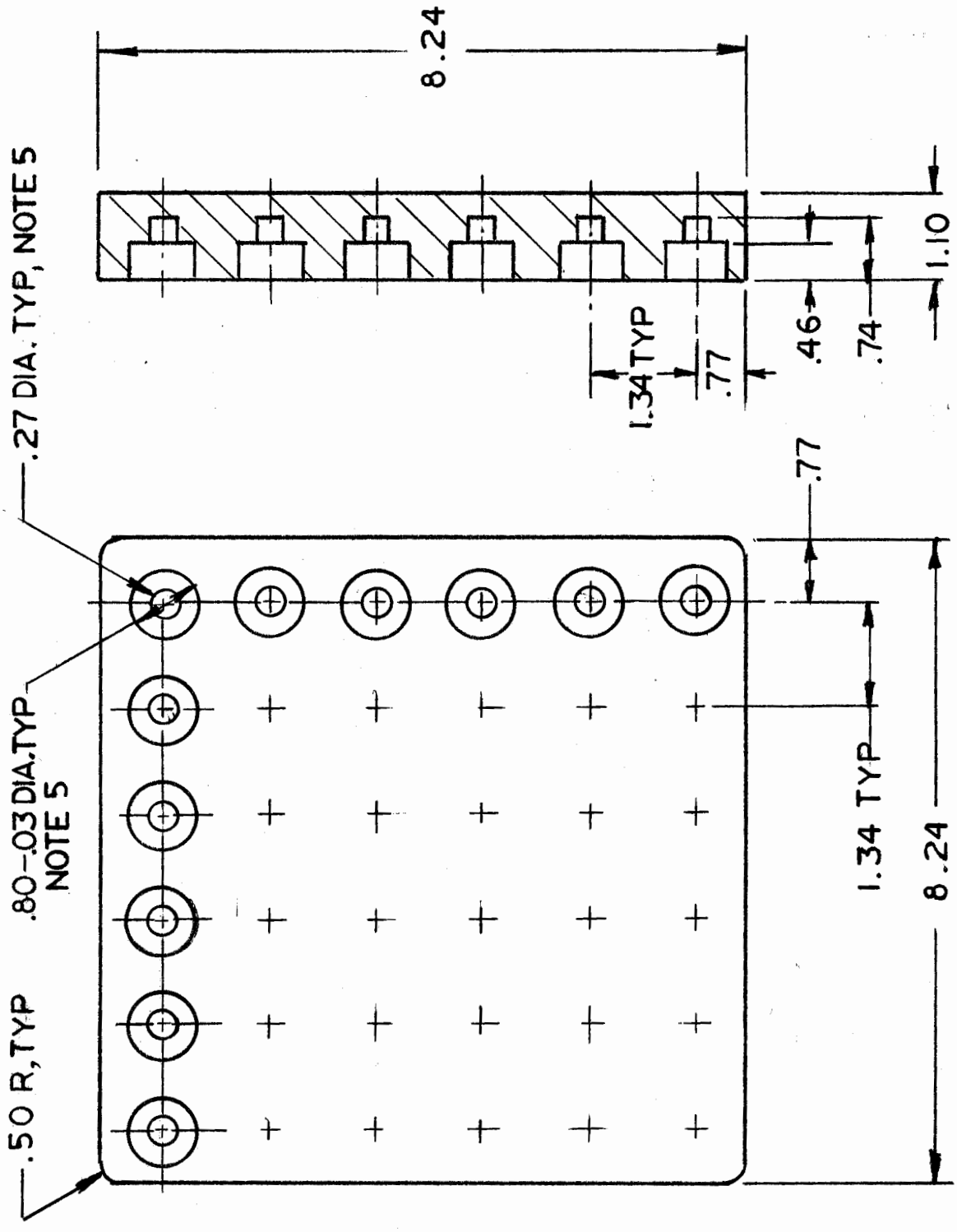
Custodian:
ARMY-PA

Preparing Activity:
ARMY-PA

Project Number: FSC: 1390-A073

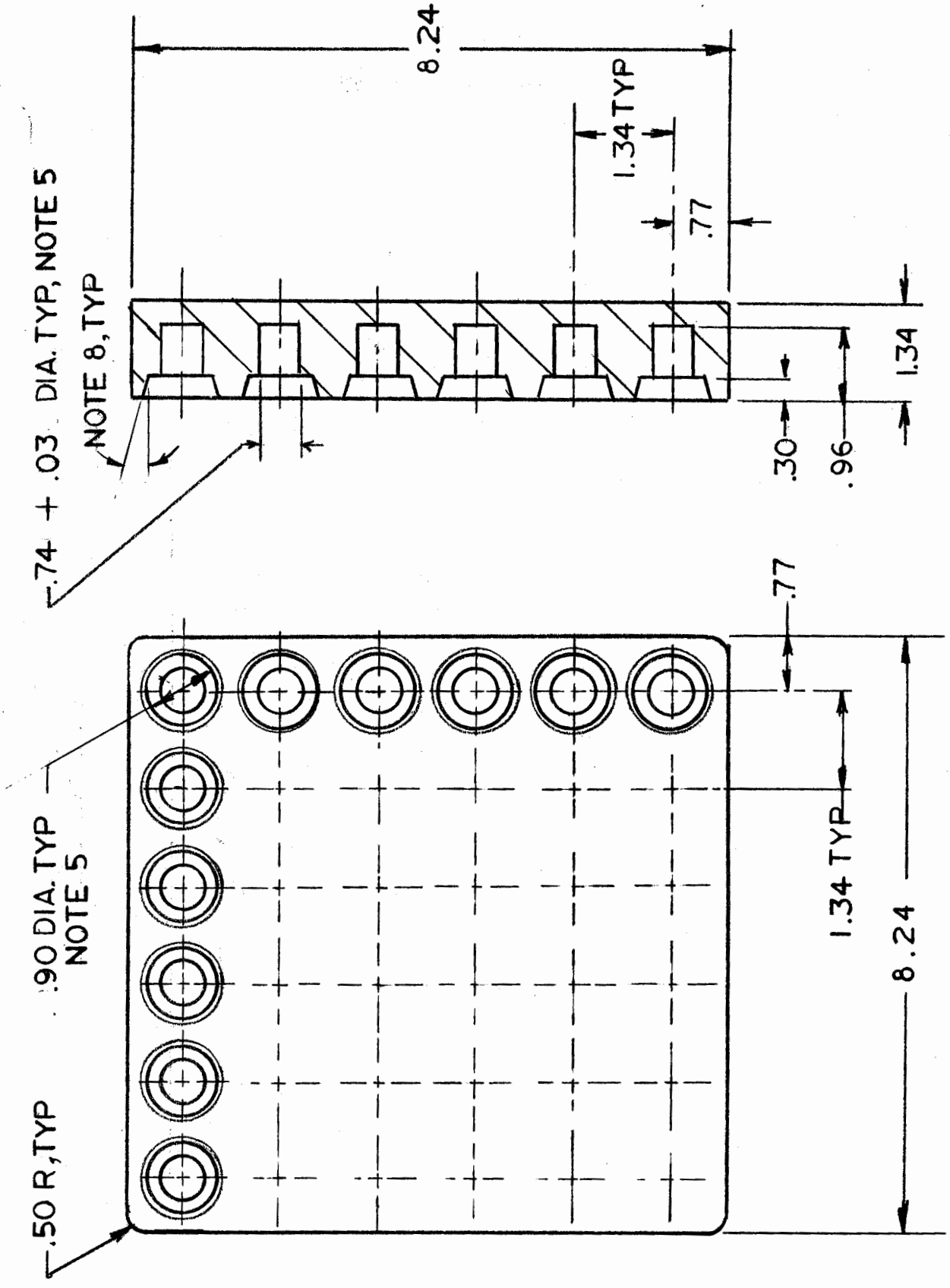
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

FIGURE 1.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

FIGURE 2



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.

NOTES

1. Spec. MIL-A-2550 applies.
2. Material: Plastic Foam, Polystyrene, Type II, Class 2, MIL-P-60312, Note 3.
3. Foam density: 2 lb/ft.^{3}.
4. Draft angle on perimeter of tray to be 3° max.
5. Dimension corresponds to surface shown. Top of cavity to be larger because of draft angle.
6. All unspecified draft angles to be 2° max.
7. All untoleranced dimensions $\pm$ .03.
8. Draft angle on this surface to be 12 - 15°.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:38Z
Check the source to verify that this is the current version before use.